

B-5-A

Roll No.....

Total No. of Questions : 26]

[Total No. of Printed Pages : 8

XIISZRJDF20**1105-A****MATHEMATICS****Time : 3 Hours]****[Maximum Marks : 100****(Long Answer Type Questions)****6 each**

1. Using properties of determinants, show that :

$$\begin{vmatrix} x+y+2z & x & y \\ z & y+z+2x & y \\ z & x & z+x+2y \end{vmatrix} = 2(x+y+z)^3$$

Or

Solve the following system of equations by matrix method :

$$3x - 2y + 3z = 8$$

$$2x + y - z = 1$$

$$4x - 3y + 2z = 4$$

XIISZRJDF20-1105-A**Turn Over****B-5-A**

(2)

2. Differentiate the function

$$x^{\sin x} + (\sin x)^{\cos x}$$

Or

If $y = 3 \cos (\log x) + 4 \sin (\log x)$, show that :

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$$

3. Evaluate :

$$\int \frac{\cos x \, dx}{(1 - \sin x)(2 - \sin x)}$$

Or

Find $\int_0^2 (x^2 + 1) \, dx$ as the limit of a sum.

4. Show that the differential equation $(x^2 + xy) \, dy = (x^2 + y^2) \, dx$ is homogeneous and solve it.

Or

Find the general solution of the differential equation :

$$x \frac{dy}{dx} + 2y = x^2 \log x$$

XIISZRJDF20—1105-A

B-5-A

(3)

5. Find the shortest distance between the lines :

$$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$$

and

$$\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$$

Or

Find the angle between the planes whose vector equations are :

$$\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) = 5$$

and

$$\vec{r} \cdot (3\hat{i} - 3\hat{j} + 5\hat{k}) = 3$$

(Short Answer Type Questions)

4 each

6. Find the value of K so that the function "f" defined by :

$$f(x) = \begin{cases} Kx + 1, & \text{if } x \leq 5 \\ 3x - 5, & \text{if } x > 5 \end{cases}$$

is continuous at $x = 5$.

7. If $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$, show that $f \circ f(x) = x$, for all $x \neq \frac{2}{3}$. What

is the inverse of 'f' ?

XIISZRJDF20—1105-A

B-5-A

Turn Over

8. If $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$, then find the value of x .
9. Using the properties of determinants and without expanding, prove that :

$$\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0$$

10. A bag contains 4 red and 4 black balls, another bag contains 2 red and 6 black balls. One of the two bags is selected at random and a ball is drawn from the bag which is found to be red. Find the probability that the ball is drawn from the first bag.
11. If A and B are two independent events, then the probability of occurrence of at least one of A and B is given by :

$$1 - P(A') P(B')$$

Prove it.

12. Find the vector equation of a plane which is at a distance of 7 units from the origin and normal to the vector $3\hat{i} + 5\hat{j} - 6\hat{k}$.

XIISRJDF20—1105-A

B-5-A

(5)

13. Find $|\vec{a} \times \vec{b}|$, if $\vec{a} = \hat{i} - 7\hat{j} + 7\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} + 2\hat{k}$.

14. Evaluate :

$$\int_0^1 \frac{x dx}{x^2 + 1}$$

15. Solve the following linear programming problems graphically :

$$\text{Maximise :} \quad z = 5x + 3y$$

Subject to :

$$3x + 5y \leq 15$$

$$5x + 2y \leq 10$$

$$x \geq 0, y \geq 0$$

(Very Short Answer Type Questions)

2 each

16. Express $\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right)$, $-\frac{\pi}{2} < x < \frac{3\pi}{2}$ in the simplest form.

17. Find x and y , if :

$$2\begin{pmatrix} 1 & 3 \\ 0 & x \end{pmatrix} + \begin{pmatrix} y & 0 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 6 \\ 1 & 8 \end{pmatrix}$$

XIISZRJDF20-1105-A

B-5-A

Turn Over

18. Find $\frac{dy}{dx}$, if $2x + 3y = \sin x$.
19. Find $\frac{dy}{dx}$, if $x = 2at^2$, $y = at^4$.
20. The volume of a cube is increasing at the rate of $8 \text{ cm}^3/\text{s}$. How fast is the surface area increasing when the length of an edge is 12 cm ?
21. Find the angle between the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$.
22. Shade the feasible region of L.P.P. $x + 3y \geq 3$, $x + y \geq 2$, $x, y \geq 0$.
23. Compute $P(A/B)$, if $P(B) = 0.5$ and $P(A \cap B) = 0.32$.
24. Evaluate :

$$\int x \sin x \, dx$$

25. Evaluate :

$$\int_0^1 \frac{dx}{1+x^2}$$

(Objective Type Questions)

1 each

26. (i) Define Identity Matrix.
- (ii) Area of the region bounded by the curve $y^2 = 4x$, y -axis and the line $y = 3$ is $\frac{9}{4}$. (True/False)

XIISZRJDF20-1105-A

B-5-A

(7)

(iii) The degree of the differential equation $\frac{dy}{dx} - \cos x = 0$ is 0.

(True/False)

(iv) $\frac{d}{dx}(\sin^2 x + \cos^2 x) = \dots\dots\dots$ (Fill in the blank)

(v) $\int \sqrt{a^2 + x^2} dx = \dots\dots\dots$ (Fill in the blank)

(vi) The direction cosines of x-axis are $\dots\dots\dots$.

(Fill in the blank)

(vii) The rate of change of the area of a circle with respect to its radius 'r', when $r = 5$ cm is :

(A) $8\pi \text{ cm}^2/\text{cm}$

(B) $10\pi \text{ cm}^2/\text{cm}$

(C) $12\pi \text{ cm}^2/\text{cm}$

(D) None of these

(viii) The slope of tangent to the curve $y = x^3 - x$ at $x = 2$ is :

(A) 11

(B) 12

(C) 13

(D) None of these

(ix) $\int x^2 e^{x^3} dx$ is equal to :

(A) $\frac{1}{3} e^{x^3} + C$

(B) $\frac{1}{3} e^{x^2} + C$

(C) $\frac{1}{2} e^{x^3} + C$

(D) $\frac{1}{2} e^{x^2} + C$

(x) Integrating factor of the differential equation $x \frac{dy}{dx} + 2y = x^2$ is

(A) x

(B) x^2

(C) x^3

(D) None of these